

Geometry of Recursive Knots:

Iterated Cables by Frame Offsets, Their Thickness and Ropelength

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Abstract

We study a concrete family of *recursive knots*: starting from a round circle K_0 , the curve K_d is the (p_d, q_d) torus pattern placed on the tube of radius r_d around K_{d-1} , using a rotation-minimising (Bishop) frame closed by a linear twist. The question of the research line is how much rope (length divided by thickness) such nested knots need. We prove elementary facts for this family: K_d is an embedded smooth closed curve whenever $r_d < \text{thick}(K_{d-1})$; the closed Bishop frame has linking number $\text{Wr } K_{d-1} - \alpha_d/2\pi$ with its core, which identifies K_d as an explicit cable of K_{d-1} ; $L(K_d)$ lies between $p_d L(K_{d-1})(1 - r_d \kappa_{\max}(K_{d-1}))$ and $p_d L(K_{d-1})(1 + r_d \kappa_{\max}(K_{d-1})) + r_d(2\pi q_d + p_d |\alpha_d|)$; and, for $p_d = 2$, $\text{thick}(K_d) \leq r_d$, so that $\text{Rop}(K_d) \geq L(K_d)/r_d$ inside the family. For the lower bound $\text{thick}(K_d) \geq \min(\text{thick } K_{d-1} - r_d, c r_d \sin(\pi/p_d))$ we treat $p_d = 2$, $r_d \leq \text{thick}(K_{d-1})/2$: in the default family $(2, 3)$ every doubly critical pair of K_d with distinct base points, on different strands or on the same strand (strands being defined relative to the shorter base arc), is at distance $\geq \min(2(\text{thick } K_{d-1} - r_d), r_d)$, at every depth and without further hypotheses (Lemma 3.7, Corollary 3.12; the same-strand case was obtained in the second round of review). The hypothesis (H_c) with $c = 1/2$ therefore reduces to the curvature bound $\min \text{Rad}(K_d) \geq r_d/2$. That bound is proved at depth one, so $\text{thick}(K_1) \geq r_1/2$ for every $f = r_1 \leq 1/2$; for $d \geq 2$ it follows from an explicit estimate under an additional bound (H_3) on the derivatives of the speed and of the parallel-frame curvature vector of K_{d-1} , whose constants we only evaluate on grids with inputs measured on the polygons (uncertified), and without such a bound no constant works for an arbitrary base (Remark 3.15). Under (H_c) the ropelength bound satisfies a linear recursion, $\text{Rop}_d \leq 2\pi \Lambda^d$ with an explicit Λ . Numerically, explicit polygons with up to 8192 vertices give $L/\tau = 38.4, 155.6$ and 622.7 at depths 1, 2, 3 for the default pattern $(2, 3)$ and offset fraction $f = 1/2$, stable to 0.03% under doubling of the resolution. These are numerically verified upper bounds for explicit polygons, not certified smooth thicknesses; no stationarity, optimality or recursive flow law is claimed.

1 Introduction and status

The research line. Line RTK001 (“Geometry of recursive knots”) asks how nested knots—knots tied on the tube of other knots, recursively—can be represented as tubes without self-intersection, and how much length that costs. Its public record states one finding (a feasible upper bound at depth one) and an explicit scope restriction: neither stationarity nor optimality of any configuration is certified, and no general recursive flow law has been established. No manuscript of the line existed before this draft; the family below was fixed from the record. Version 0.2 incorporated a first round of internal review (30/09/2026): the thickness hypothesis of v0.1 was false as stated for an arbitrary base curve (Remark 3.15) and became a statement about the family, partially proved. Version 0.3 incorporates a second round (03/10/2026): the status of every constant is now labelled as analytic or as a grid evaluation, and the same-strand case, open in v0.2, is proved (Proposition 3.11), which proves (H_c) at depth one and reduces it to a curvature bound at higher depth (Theorem 3.14).

What is done here. We (i) fix a definite family of recursive knots (Section 2); (ii) prove what is elementary about it—embeddedness, the framing it uses, two-sided length bounds, an upper bound on the thickness—and prove part of the thickness lower bound, isolating in one hypothesis on the family what is not proved (Section 3); (iii) compute explicit polygonal configurations at depths 1 to 3, their polygonal thickness and ropelength, with a resolution check (Section 4); (iv) tabulate every claim with its

status (Section 5). Lower bounds for general knots are only quoted from the literature (Remark 3.19); the only lower bound we derive is internal to the family (Proposition 3.5).

2 Setting

2.1 Thickness and ropelength

Let $K \subset \mathbb{R}^3$ be a closed $C^{1,1}$ curve of length $L(K)$. Its *thickness* $\text{thick}(K)$ is the supremum of ρ such that the open tube of radius ρ about K is embedded (normal injectivity radius). Two equivalent descriptions are classical. Litherland, Simon, Durumeric and Rawdon [11] show, for C^2 curves (extended to the $C^{1,1}$ case in [3]; all curves K_d below are C^∞),

$$\text{thick}(K) = \min(\min\text{Rad}(K), \tfrac{1}{2} \text{dcsd}(K)), \quad (1)$$

where $\min\text{Rad}$ is the minimal radius of curvature and dcsd the *doubly critical self distance*, the minimal distance between pairs $x \neq y$ of K whose chord is perpendicular to K at both ends. Gonzalez and Maddocks [7] show $\text{thick}(K) = \inf_{x \neq y} \rho_{pt}(x, y)$, with $\rho_{pt}(x, y) = |x - y| / (2 \sin \angle(T_x, y - x))$ the radius of the circle through y tangent to K at x . The *ropelength* is $\text{Rop}(K) = L(K) / \text{thick}(K)$; it is scale invariant, $\text{Rop}(K) \geq 2\pi$ for every closed curve (Fenchel's theorem [5], total curvature $\geq 2\pi$, together with $\kappa \leq 1/\text{thick}$), with equality only for the circle, and $\text{thick}(K) \leq \min\text{Rad}(K)$ always.

2.2 Frames, holonomy and the closing twist

Let K be a closed C^2 regular curve parametrised on $[0, 2\pi)$ (not necessarily by arclength), with unit tangent T and curvature κ . A *rotation-minimising* or Bishop frame [1] is an orthonormal frame (T, N_0, B_0) with $N'_0 = -\kappa_1 |K'| T$, $B'_0 = -\kappa_2 |K'| T$, $\kappa_1^2 + \kappa_2^2 = \kappa^2$; it is unique up to a constant rotation in the normal plane and is in general *not* periodic: after one turn the transported normal is rotated by a holonomy angle $\alpha \in (-\pi, \pi]$. We close it by the linear twist $\omega(t) = -\alpha t / 2\pi$, i.e. $N = \cos \omega N_0 + \sin \omega B_0$, $B = T \times N$. The closed frame satisfies

$$N' = -\tilde{\kappa}_1 |K'| T + \omega' B, \quad B' = -\tilde{\kappa}_2 |K'| T - \omega' N, \quad |\omega'| = \frac{|\alpha|}{2\pi} \leq \frac{1}{2}, \quad (2)$$

with the rotated curvature components $\tilde{\kappa}_1 = \kappa_1 \cos \omega + \kappa_2 \sin \omega$, $\tilde{\kappa}_2 = -\kappa_1 \sin \omega + \kappa_2 \cos \omega$ (same norm κ); below we write κ_1, κ_2 for whichever frame is in use.

2.3 The family of recursive knots

Definition 2.1 (iterated cable by frame offsets). Let K_0 be the circle of radius $R_0 = 1$. For $d \geq 1$ choose coprime integers $p_d \geq 2$, $q_d \geq 1$, a radius $r_d > 0$ and a phase φ_d . Reparametrise K_{d-1} to period 2π by the linear change $t \mapsto t/p_{d-1}$ (so that K_{d-1} is traversed at parameter speed $|K'_{d-1}|$, not at unit speed), let (N_{d-1}, B_{d-1}) be its closed Bishop frame, and set for $t \in [0, 2\pi p_d)$

$$K_d(t) = K_{d-1}(t \bmod 2\pi) + r_d [\cos \theta_d(t) N_{d-1}(t) + \sin \theta_d(t) B_{d-1}(t)], \quad \theta_d(t) = \frac{q_d}{p_d} t + \varphi_d. \quad (3)$$

The default is $(p_d, q_d) = (2, 3)$, $\varphi_d = 0$ at every depth; r_d is a fraction f of the thickness of K_{d-1} (the guaranteed lower bound ρ_{d-1} in the theory, the measured polygonal value in the experiments).

K_1 is the (p_1, q_1) torus knot on the round torus of radii $(1, r_1)$; the knot type of K_d for $d \geq 2$ depends on the framing, which Lemma 3.2 makes explicit. The pattern angle advances in the parameter t , not in arclength; since the parametrisation of K_{d-1} is the one inherited from the construction, the *pitch* of the strands relative to the base (the quantity h of Section 3) is a property of the family, not a free parameter.

2.4 Polygonal measurements

Numerically a curve is a closed polygon $P = (x_0, \dots, x_{N-1})$. We use $L(P) = \sum |x_{i+1} - x_i|$; $\min\text{Rad}(P)$, the minimum circumradius of consecutive triples; $\text{dcsd}(P)$, the minimal distance $|x_i - x_j|$ over *doubly critical vertex pairs*, i.e. pairs where $|x_i - x_j|$ is a local minimum in i and in j (the diagonal $D_{ii} = 0$ is kept in the neighbourhood test, which excludes near-diagonal pairs); and the proxy

$$\tau(P) = \min(\min\text{Rad}(P), \tfrac{1}{2} \text{dcsd}(P)), \quad \text{Rop}(P) = L(P) / \tau(P). \quad (4)$$

The local-minimum test could miss doubly critical pairs of saddle type; as a safeguard we also compute $\tau_{pt}(P) = \min_{i \neq j} |x_j - x_i|^2 / (2|T_i \times (x_j - x_i)|)$ with central-difference tangents T_i , which uses no criticality test. Neither τ nor τ_{pt} is a certified thickness of a smooth curve (for the thickness of polygons themselves see [13]); they are the quantities of (1) evaluated on an explicit polygon, and their stability under doubling of N is the main evidence offered that they approximate the smooth values. Vertex pairs may overestimate the polygon–polygon distance; the minimal segment–segment distance over pairs of edges with cyclic index separation $> N/10$, divided by the vertex dcscd, is 0.99994, 1.00000 ($d = 1, f = 0.25, 0.5$) and 0.99993 ($d = 2, f = 0.5$), and near the critical pair the ratio is ≥ 0.99998 in every level examined (`experiments/aux_checks.py`). Writhe is the Gauss double sum over pairs of edges (midpoint rule); its accuracy is discussed in Section 4.

3 Results

Throughout, $K = K_{d-1}$ is a closed C^∞ regular curve of thickness $\tau = \text{thick}(K) > 0$, hence of maximal curvature $\kappa_{\max} \leq 1/\tau$, carrying the parametrisation of period 2π inherited from Definition 2.1, with speed $v = |K'| \in [v_{\min}, v_{\max}]$; K_d is built by (3) with $r = r_d, p = p_d, q = q_d$, and $f = r/\tau$.

Lemma 3.1 (embedding). *[proved here] If $r < \tau$ then K_d is a regular C^∞ closed curve, injective on $[0, 2\pi p)$, contained in the torus $\partial\{x : \text{dist}(x, K) < r\}$.*

Proof. Smoothness: the Bishop frame of a C^∞ curve is C^∞ (it solves a linear ODE with smooth coefficients), and (3) is a smooth expression in it; periodicity of K_d in t with period $2\pi p$ holds because $\theta_d(t + 2\pi p) = \theta_d(t) + 2\pi q$ and the closed frame has period 2π . Regularity: differentiating (3) with (2),

$$K'_d = |K'| (1 - r(\kappa_1 \cos \theta + \kappa_2 \sin \theta)) T + r(\theta' + \omega')(-\sin \theta N + \cos \theta B), \quad (5)$$

and $|\kappa_1 \cos \theta + \kappa_2 \sin \theta| \leq \kappa \leq 1/\tau$, so the tangential coefficient is at least $|K'|(1 - r/\tau) > 0$. Injectivity: let $t_1 \neq t_2$ in $[0, 2\pi p)$. If $t_1 \not\equiv t_2 \pmod{2\pi}$, the two points lie in the normal discs of radius $r < \tau$ at distinct points of K ; by definition of thickness these discs are disjoint, so the points differ. If $t_1 \equiv t_2 \pmod{2\pi}$, then $t_1 - t_2 = 2\pi k$ with $0 < |k| < p$ and $\theta_d(t_1) - \theta_d(t_2) = 2\pi qk/p \notin 2\pi\mathbb{Z}$ because $\gcd(p, q) = 1$; the two points lie in the same normal disc at the same radius $r > 0$ but at different angles, hence differ. Containment in the torus is immediate from (3). $\square$

The constant $1/2$ in $r \leq \tau/2$, used from Lemma 3.3 on, is what the estimates below need; embeddedness only needs $r < \tau$.

Lemma 3.2 (the framing used, and the knot type). *[proved here, from classical results] Let $\alpha \in (-\pi, \pi]$ be the holonomy of the Bishop frame of K . Then the closed frame (N, B) is a framing of K with linking number $n := \text{Lk} = \text{Wr } K - \alpha/2\pi \in \mathbb{Z}$; in particular $\alpha \equiv 2\pi \text{Wr } K \pmod{2\pi}$, and $n = \text{round}(\text{Wr } K)$ whenever $\text{Wr } K \notin \mathbb{Z} + \frac{1}{2}$ (if $\text{Wr } K \in \mathbb{Z} + \frac{1}{2}$ then $\alpha = \pi$ and $n = \text{Wr } K - \frac{1}{2}$). With the convention that a (p, q') -cable of K winds p times along K and q' times around it relative to a framing of linking number 0 (the Seifert framing), K_d is the $(p, q + pn)$ -cable of K [10]. In particular K_1 is the torus knot $T(p_1, q_1)$.*

Proof. By the Călugăreanu–White–Fuller formula [4, 6, 14], $\text{Lk} = \text{Tw} + \text{Wr}$ for any framing. The Bishop frame has zero twist density; the closing rotation ω adds total twist $\text{Tw} = \frac{1}{2\pi} \int \omega' dt = -\alpha/2\pi \in [-\frac{1}{2}, \frac{1}{2})$. Hence $\text{Lk} = \text{Wr} - \alpha/2\pi$, and since $\text{Lk} \in \mathbb{Z}$ the congruence and the rounding rule follow. The pattern (3) winds q times around the meridian relative to the frame (N, B) while going p times along K ; relative to a framing of linking number 0 it winds $q + pn$ times, which is the stated cable. For K_0 a planar circle, $\text{Wr} = 0 = \alpha$ and K_1 is the standard torus knot. $\square$

Lemma 3.3 (length, two-sided). *[proved here] If $r \leq \tau/2$ then*

$$p L(K) (1 - r\kappa_{\max}) \leq L(K_d) \leq p L(K) (1 + r\kappa_{\max}) + r(2\pi q + p|\alpha|) \leq p L(K) (1 + r/\tau) + r(2\pi q + \pi p). \quad (6)$$

Proof. From (5) and the triangle inequality, $|K'_d| \leq |K'| (1 + r\kappa) + r(|\theta'| + |\omega'|) = |K'| (1 + r\kappa) + r(q/p + |\alpha|/2\pi)$. Integrate over $[0, 2\pi p)$: the first term gives $p \int_0^{2\pi} |K'| (1 + r\kappa) dt \leq p L(K) (1 + r\kappa_{\max})$, the second $r(2\pi q + p|\alpha|)$. Then use $\kappa_{\max} \leq 1/\tau$ and $|\alpha| \leq \pi$. For the lower bound, $|K'_d|$ is at least its tangential component $|K'| (1 - r\kappa) \geq |K'| (1 - r\kappa_{\max})$; integrate. $\square$

Remark 3.4. The term $p|\alpha|r$ from the closing twist is absent from a naive count “ p copies plus q meridians”; it is at most πpr and is needed for the proof via (5) because the Bishop frame of a non-planar K does not close. Numerically the upper bound without it also holds in all 36 measured levels (minimal ratio bound/ L equal to 1.255), so the term is a proof device, not a numerical necessity.

Proposition 3.5 (upper bound on the thickness and a lower bound on the ropelength, $p = 2$). *[proved here] Let $p = 2$ (q odd) and $r < \tau$. Then $\text{thick}(K_d) \leq r$, hence $\text{Rop}(K_d) \geq L(K_d)/r \geq pL(K)(1 - r\kappa_{\max})/r$. In the family with $r_d = f \text{thick}(K_{d-1})$ and $f \leq 1/2$ this gives $\text{Rop}(K_d) \geq \frac{p(1-f)}{f} \text{Rop}(K_{d-1})$, so $\text{Rop}(K_d) \geq 2\pi(p(1-f)/f)^d$.*

Proof. The two points $K_d(t)$ and $K_d(t + 2\pi)$ lie in the same normal disc of K at antipodal angles (θ and $\theta + \pi q$, q odd), so their chord is $-2r(\cos \theta N + \sin \theta B)$, orthogonal to T and to $-\sin \theta N + \cos \theta B$, hence by (5) orthogonal to K'_d at both ends: the pair is doubly critical at distance $2r$, so $\text{dcsd}(K_d) \leq 2r$ and $\text{thick}(K_d) \leq r$ by (1) (equivalently, $\rho_{pt} = r$ for this pair). The rest is Lemma 3.3 with $\kappa_{\max} \leq 1/\tau$, $r = f\tau$ and $\text{Rop}(K_{d-1}) = L(K_{d-1})/\tau$. $\square$

For the default family at $f = 1/2$ the internal lower bound is $2\pi \cdot 2^d$, i.e. 12.6, 25.1, 50.3 at $d = 1, 2, 3$, against the measured 38.4, 155.6, 622.7; with the measured $L(K_d)$ the bound $L(K_d)/r_d$ is attained whenever $\text{thick}(K_d) = r_d$ (Section 4). The lower bound of (6) holds in all 36 measured levels (36 of 36).

Thickness of K_d : what is proved and what is not. Write $s = 2r \sin(\pi/p)$ for the chord between adjacent strands in a normal disc of K (the p points $K_d(t + 2\pi k)$, $k = 0, \dots, p-1$, sit at the p equally spaced angles $\theta + 2\pi qk/p$, since $\gcd(p, q) = 1$).

Proposition 3.6 (two of the three cases). *[proved here] If $r \leq \tau/2$: (a) two points of K_d in the same normal disc of K are at distance $\geq s$; (b) two points of K_d whose base points on K are at distance $\geq 2\tau$ are at distance $\geq 2(\tau - r)$.*

Proof. (a) They are at the same radius r and at angles differing by $2\pi qk/p$, $0 < k < p$, so their distance is $2r|\sin(\pi qk/p)| \geq 2r \sin(\pi/p) = s$ because $qk \not\equiv 0 \pmod{p}$. (b) Triangle inequality: $|K_d(t_1) - K_d(t_2)| \geq |K(t_1) - K(t_2)| - 2r$. $\square$

Not covered by Proposition 3.6 are the pairs with $0 < |K(t_1) - K(t_2)| < 2\tau$, and the curvature of K_d . These include nearby base points (the helix geometry of p strands on a bent tube) and base points on different arcs of K , e.g. across the hole of a torus: at $f = 1/2$, $d = 1$ the doubly critical vertex pair that realises dcsd has base points 103.7° apart on K_0 , at distance $1.573 < 2\tau_0 = 2$ (`aux_checks.py`), while for $f \leq 0.35$ it is the antipodal pair of one normal disc. For a straight tube the p strands are helices with pitch parameter $h = |K'|p/q$ and the antipodal pair ($p = 2$) is a local minimum of the strand distance exactly when $h \geq r$; at depth one the relevant pitch on the inner equator is $(1-r)p/q$, and $(1-r)p/q \geq r$ iff $r \leq p/(p+q) = 0.4$ for $(2, 3)$, consistent with $\text{thick}(K_1) = r_1$ for $f \leq 0.35$ and $\text{thick}(K_1) < r_1$ for $f = 0.5$.

Notation for the local case. Let (T, N_0, B_0) be the (non-periodic) Bishop frame of K and α its holonomy, and set

$$m := \frac{q}{p} - \frac{\alpha}{2\pi}, \quad h_{\min} := \frac{v_{\min}}{m}, \quad \Theta_{\min} := \frac{h_{\min}}{\tau}, \quad \zeta_{\min} := \frac{(1-f)h_{\min}}{r}, \quad f = \frac{r}{\tau} \leq \frac{1}{2}.$$

In the parallel frame the strand of (3) turns at the constant rate m (its angle is $\psi = \theta + \omega = mt + \varphi$), so $h = v/m$ is the pitch parameter above ($|K'|p/q$ when $\alpha = 0$) and $\Theta_{\min}$ is the smallest pitch in units of the base thickness. For two points $X_i = K(t_i) + rU_i$ of K_d with distinct base points, let σ be the length of the *shorter* base arc between $K(t_1)$ and $K(t_2)$, $\delta = \sigma/\tau$, s the base-parameter difference along that arc and $u = m|s|$ the relative turning of the strands along it. The integer $k = (t_2 - t_1 - s)/2\pi \pmod{p}$ is the strand index of X_2 relative to X_1 *along that arc* (for $p = 2$ the two strands are exchanged by one turn around K , so “same strand” is meaningful only relative to a base arc). With $\ell_0(\delta) = 2\tau \sin(\delta/2)$, $E(\delta) = \tau(1 - \cos \delta)$ for $\delta \leq \pi/2$ and $E(\delta) = \tau(1 + \delta - \pi/2)$ for $\delta > \pi/2$, $\beta(\delta) = \delta^2/(4 \sin(\delta/2))$, define

$$\Lambda(\delta) = \ell_0 - \frac{2rE}{\ell_0}, \quad P_1(\delta) = \left[2 \cos\left(\frac{\delta}{2\Theta_{\min}}\right) \sqrt{1 - \beta(\delta)^2} - \frac{\delta^2}{3} \right]_+ \quad (7)$$

(with $P_1 := 0$ if $\delta > \pi\Theta_{\min}$ or $\beta > 1$), and

$$c_1(f, \Theta_{\min}) := \frac{1}{2r} \inf_{0 < \delta < \pi} \max \left\{ \sqrt{\Lambda(\delta)_+^2 + r^2 P_1(\delta)^2}, \ell_0(\delta) - 2r \right\}. \quad (8)$$

All lengths in (8) scale with r , so c_1 depends only on $(f, \Theta_{\min})$; $\ell_0/r - 2$ and $\Lambda/r = (2/f) \sin(\delta/2) - (E/\tau)/\sin(\delta/2)$ decrease in f and P_1 does not depend on f and increases in $\Theta_{\min}$, so c_1 is non-increasing in f and non-decreasing in $\Theta_{\min}$.

Lemma 3.7 (pairs on different strands, $p = 2$). *[proved here] Let $p = 2$, q odd, $r \leq \tau/2$. Any two points X_1, X_2 of K_d with distinct base points and strand index $k = 1$ relative to the shorter base arc satisfy*

$$|X_2 - X_1| \geq \min(2(\tau - r), 2c_1(f, \Theta_{\min})r).$$

Moreover $c_1(\frac{1}{2}, \frac{2}{3}) \geq \frac{1}{2}$ (a grid evaluation gives 0.6117), hence $c_1 \geq \frac{1}{2}$ whenever $f \leq \frac{1}{2}$ and $\Theta_{\min} \geq \frac{2}{3}$. For the default family $(p, q) = (2, 3)$ with $r_j \leq \text{thick}(K_{j-1})/2$ for all $j \leq d$ one has $\Theta_{\min} \geq \frac{2}{3}$ at every depth, so every such pair is at distance $\geq \min(2(\tau - r), r)$.

Proof. Step 1 (chord of a thick curve). Let $x = K(0)$ and $g(\sigma) = |K(\sigma) - x|$ in arclength. A local minimum y of g in $(0, L)$ has $(y - x) \perp T(y)$, hence $\rho_{pt}(y, x) = |x - y|/2 \geq \tau$, so $|x - y| \geq 2\tau$. Consequently $\{g < 2\tau\}$ consists of an initial interval $[0, a)$ and a final one $(L - b, L)$ on which g is monotone (a non-monotone stretch below 2τ would contain an interior local minimum below 2τ), and on them $g' = \cos \angle(T, K - x) \geq \sqrt{1 - g^2/4\tau^2}$ a.e. (from $\rho_{pt}(K(\sigma), x) \geq \tau$), which integrates to $g(\sigma) \geq 2\tau \sin(\sigma/2\tau)$ and shows $a, b \leq \pi\tau$ (this is the classical chord-arc inequality for thick curves, cf. [3, 7]). Hence for base points with shorter arc $\sigma = \tau\delta \leq L/2$: either $\delta \geq \pi$ and $|x - y| \geq 2\tau$ (then Proposition 3.6(b) gives $|X_2 - X_1| \geq 2(\tau - r)$), or $\delta < \pi$ and $\ell := |x - y| \geq \ell_0(\delta)$.

Step 2 (elementary estimates, $\delta < \pi$). Let $y = K(t_2)$, $e = (y - x)/\ell$ and $\bar{T} = \sigma^{-1} \int vT dt$ over the arc, so $e = \bar{T}/|\bar{T}|$, $|\bar{T}| = \ell/\sigma$. Since $|dT/d\sigma| = \kappa \leq 1/\tau$: (E2) $|\bar{T} - T(x)| \leq \delta/2$; (E3) for $w \perp T(x)$, $|w \cdot e| = |w \cdot \Pi_x e| \leq |w| |\bar{T} - T(x)| \sigma/\ell \leq |w| \beta(\delta)$, where Π_x is the projection onto the normal plane at x ; (E4) for unit $w \perp T(x)$, $|w \cdot (y - x)| = |\int v w \cdot \Pi_x T dt| \leq \int_0^\sigma \sin \min(\sigma'/\tau, \pi/2) d\sigma' = E(\delta)$, and likewise for $w \perp T(y)$; (E5) if $\tilde{U}_2$ denotes the vector with the same parallel-frame angle as U_2 but in the normal plane at x , then $U_2 - \tilde{U}_2 = -\int \kappa_\perp vT dt$ with $|\kappa_\perp| \leq \kappa$ (differentiate $\cos \psi_2 N_0 + \sin \psi_2 B_0$ along the arc), so $|U_2 - \tilde{U}_2| \leq \delta$ and its component orthogonal to e is at most $\int \kappa v |T - \bar{T}| dt \leq \int_0^\sigma \frac{\sigma'^2 + (\sigma - \sigma')^2}{2\sigma\tau^2} d\sigma' = \delta^2/3$; (E6) $\tilde{U}_2 - U_1$ lies in the normal plane at x and has length $2|\sin(\Delta\psi/2)|$, where $\Delta\psi = ms + 2\pi qk/p$ is the relative angle of the strands after parallel transport along the arc (the holonomy terms cancel because $N_0(t + 2\pi)$ is $N_0(t)$ rotated by α and m contains $-\alpha/2\pi$), i.e. $|\sin(\Delta\psi/2)| = |\cos(u/2)|$ for $k = 1$ and $p = 2$.

Step 3 (decomposition). Write $X_2 - X_1 = (y - x) + r(U_2 - U_1) = Ae + Q$ with $Q \perp e$. By (E4), $A = \ell + r(U_2 - U_1) \cdot e \geq \ell - 2rE/\ell \geq \Lambda(\delta)$, since $\ell \mapsto \ell - 2rE/\ell$ is increasing and $\ell \geq \ell_0$. By (E3), (E5), (E6), $|Q| = r|(U_2 - U_1)_\perp| \geq r[2|\cos(u/2)|\sqrt{1 - \beta^2} - \delta^2/3]_+$. Finally $u = m|s| \leq m\sigma/v_{\min} = \delta/\Theta_{\min}$ and $|\cos(u/2)|$ is decreasing on $[0, \pi]$, so $|\cos(u/2)| \geq \cos(\delta/2\Theta_{\min})$ when $\delta \leq \pi\Theta_{\min}$; together with the trivial bound $|X_2 - X_1| \geq \ell - 2r \geq \ell_0 - 2r$ this gives $|X_2 - X_1| \geq 2c_1(f, \Theta_{\min})r$ for every pair with $\delta < \pi$.

Step 4 ($c_1(\frac{1}{2}, \frac{2}{3}) \geq \frac{1}{2}$). Put $r = 1$, $\tau = 2$, $\Theta_{\min} = 2/3$; we show that the maximum in (8) is ≥ 1 for every $\delta \in (0, \pi)$. For $\delta \in [\pi/3, \pi/2]$, $\Lambda = 2(\tau - r)\sin(\delta/2) = 2\sin(\delta/2) \geq 1$. For $\delta \in [\pi/2, \pi]$, with $\varsigma = \sin(\delta/2) \in [1/\sqrt{2}, 1]$, $\Lambda = 4\varsigma - (1 + 2\arcsin \varsigma - \pi/2)/\varsigma$, and $\Lambda \geq 1$ amounts to $\phi(\varsigma) := 4\varsigma^2 - \varsigma - 1 + \pi/2 - 2\arcsin \varsigma \geq 0$; $\phi'' = 8 - 2\varsigma(1 - \varsigma^2)^{-3/2}$ changes sign once on the interval, so ϕ' is increasing then decreasing, and since $\phi'(1/\sqrt{2}) > 0$ and $\phi'(1^-) = -\infty$, ϕ increases then decreases; its minimum is at an endpoint, $\phi(1/\sqrt{2}) = 0.29$, $\phi(1) = 0.43$. For $\delta \in (0, \pi/3)$ we need $4\sin^2(\delta/2) + P_1^2 \geq 1$. On $[0.6, \pi/3]$: $4\sin^2(\delta/2) \geq 4\sin^2(0.3) > 0.349$, $\cos(3\delta/4) \geq \cos(\pi/4) > 0.7071$, $\beta \leq \beta(\pi/3) < 0.5484$ (β is increasing), so $\sqrt{1 - \beta^2} > 0.8362$, and $\delta^2/3 \leq 0.3656$; hence $P_1 > 2 \cdot 0.7071 \cdot 0.8362 - 0.3656 > 0.8169$, $P_1^2 > 0.667$, and the sum exceeds 1.01. On $(0, 0.6]$: $\cos(3\delta/4) \geq \cos(0.45) \geq 0.900$, $\beta \leq \beta(0.6) \leq 0.305$, $\delta^2/3 \leq 0.12$, so $P_1 \geq 2 \cdot 0.900 \cdot 0.952 - 0.12 \geq 1.59$. All inequalities are non-strict where needed: at $\delta = \pi/3$, $\Lambda = 1$ exactly. Hence $c_1(\frac{1}{2}, \frac{2}{3}) \geq \frac{1}{2}$. The value 0.6117 (attained near $\delta = 1.23$; an independent evaluation in the review gave 0.611895) is a grid infimum with a Lipschitz constant estimated on the grid itself (experiments/check_Hc.py), not a certified bound.

Step 5 (the family). $\Theta_{\min} = v_{\min}/(m\tau)$. By Proposition 3.5, $\tau_{d-1} = \text{thick}(K_{d-1}) \leq r_{d-1} \leq \tau_{d-2}/2$, so $\tau_{d-1} \leq 2^{-(d-1)}$. After the reparametrisation $t \mapsto t/p$, (5) gives $v_{d,\min} \geq p(1 - f)v_{d-1,\min} \geq v_{d-1,\min}$ for $p = 2$, $f \leq 1/2$, so $v_{\min} \geq v_0 = 1$ at every depth; and $m \leq q/p + 1/2 = 2$. Hence $\Theta_{\min} \geq 2^{d-2} \geq 1$ for $d \geq 2$, while for $d = 1$ ($\alpha = 0$, $m = 3/2$, $v = \tau = 1$) $\Theta_{\min} = 2/3$. Monotonicity of c_1 concludes. $\square$

Proposition 3.8 (curvature of K_d). *[proved here, under (H₃)] Assume in addition (H₃): $\sup |v'| \leq M_v$ and $\sup \sqrt{\kappa_1'^2 + \kappa_2'^2} \leq M_\kappa$ for K (derivatives in t , parallel-frame curvatures; $\kappa_1'^2 + \kappa_2'^2 = \kappa'^2 + v^2\kappa^2\tau_{\text{tors}}^2$ includes the torsion). Then, with $g(\zeta) = \zeta(\zeta^2 + 2)/(\zeta^2 + 1)^{3/2}$ and $G(\zeta_{\min}) = \sup_{\zeta \geq \zeta_{\min}} g \leq g(\sqrt{2}) = 1.0887$,*

$$\kappa_{K_d} \leq \bar{\kappa}_d := \frac{G(\zeta_{\min})}{\tau - r} + \frac{1}{r(1 + \zeta_{\min}^2)} + \frac{rm[M_v(1 + f) + r v_{\max} M_\kappa]}{(v_{\min}^2(1 - f)^2 + r^2 m^2)^{3/2}}, \quad \text{minRad}(K_d) \geq 1/\bar{\kappa}_d. \quad (9)$$

Proof. Let $U = \cos \psi N_0 + \sin \psi B_0$, $W = T \times U$, $\kappa_\perp = \kappa_1 \cos \psi + \kappa_2 \sin \psi$, $\kappa_W = -\kappa_1 \sin \psi + \kappa_2 \cos \psi$ ($\kappa_\perp^2 + \kappa_W^2 = \kappa^2$), $x = r\kappa_\perp \in [-f, f]$. Differentiating (5) with $U' = mW - \kappa_\perp vT$, $W' = -mU - \kappa_W vT$, $T' = v(\kappa_\perp U + \kappa_W W)$:

$$\begin{aligned} K'_d &= c_T T + c_W W, & K''_d &= aT + b_U U + b_W W, \\ c_T &= v(1-x), & c_W &= rm, & b_U &= v^2(1-x)\kappa_\perp - rm^2, & b_W &= v^2(1-x)\kappa_W, \end{aligned}$$

$a = a_0 - 2rv m \kappa_W$, $a_0 = v'(1-x) - rv(\kappa'_1 \cos \psi + \kappa'_2 \sin \psi)$. Since (T, U, W) is orthonormal and right-handed, $|K'_d \times K''_d|^2 = b_U^2 |K'_d|^2 + (c_W a - c_T b_W)^2$. Put $A = v^2(1-x)^2$, $B = r^2 m^2$, so $|K'_d|^2 = A + B$ and $c_W a - c_T b_W = r m a_0 - \kappa_W v(A + 2B)$. By Minkowski, $\kappa_{K_d} \leq \sqrt{v^4(1-x)^2 \kappa_\perp^2 (A+B) + \kappa_W^2 v^2 (A+2B)^2} / (A+B)^{3/2} + (rm^2(A+B)^{1/2} + rm|a_0|) / (A+B)^{3/2}$. In the first term $A(A+B) \leq (A+2B)^2$ and $\kappa_\perp^2 + \kappa_W^2 = \kappa^2$ give $\leq v\kappa(A+2B)/(A+B)^{3/2}$, which is decreasing in $A \geq v^2(1-f)^2$; substituting $A = v^2(1-f)^2$ and $\zeta = v(1-f)/(rm)$ yields $\kappa g(\zeta)/(1-f) \leq G(\zeta_{\min})/(\tau(1-f))$. The second term is at most $1/(r(1+\zeta_{\min}^2)) + rm[M_v(1+f) + rv_{\max} M_\kappa]/(v_{\min}^2(1-f)^2 + r^2 m^2)^{3/2}$. $\square$

Proposition 3.9 (pairs on the same strand, $p = 2$). *[proved here, under (H₃)] Under the hypotheses of Proposition 3.8, every doubly critical pair of K_d with distinct base points and strand index $k = 0$ relative to the shorter base arc satisfies $|X_2 - X_1| \geq \min(2(\tau - r), 2c_0 r)$, where*

$$\begin{aligned} c_0 &:= \frac{1}{2r} \inf \left\{ \max(\sqrt{\Lambda_+^2 + r^2 P_0^2}, \ell_0 - 2r) : 0 < \delta < \pi, \frac{\delta}{\Theta_{\max}} \leq u \leq \frac{\delta}{\Theta_{\min}}, u \geq m s_0 \right\}, \\ P_0 &= \left[2 \left| \sin \frac{u}{2} \right| \sqrt{1 - \beta^2 - \frac{\delta^2}{3}} \right]_+, \end{aligned}$$

$\Theta_{\max} = v_{\max}/(m\tau)$ and $s_0 = \pi/(\bar{\kappa}_d V_d)$, $V_d = v_{\max}(1+f) + rm \geq \sup |K'_d|$.

Proof. For a doubly critical pair, along the arc of K_d that follows the chosen base arc (of K_d -length $\sigma_d \leq V_d |s|$), $0 = (X_2 - X_1) \cdot T_d(X_1) = \int_0^{\sigma_d} T_d \cdot T_d(X_1) d\sigma \geq \int_0^{\sigma_d} \cos(\bar{\kappa}_d \sigma) d\sigma = \sin(\bar{\kappa}_d \sigma_d)/\bar{\kappa}_d$, which is positive unless $\sigma_d \geq \pi/\bar{\kappa}_d$; hence $|s| \geq s_0$ and $u \geq m s_0$. For $k = 0$, $\Delta\psi = \pm u$ in Step 2 of Lemma 3.7, and Step 3 gives the bound with $|\sin(u/2)|$ in place of $|\cos(u/2)|$; the infimum is over the whole admissible range of u , and the lower end $u \geq \delta/\Theta_{\max}$ (from $\sigma \leq v_{\max}|s|$) now matters because small u is the worst case. $\square$

Same-strand pairs by a turning argument (added in v0.3). Proposition 3.9 excludes short same-strand pairs through the global bound $\bar{\kappa}_d$ and the crude conversion $u \geq m s_0$; at $f = 1/2$ it gives only $c_0 = 0.32, 0.13, 0.16$ at $d = 1, 2, 3$. The following, obtained during the second round of review, measures instead the turning of the tangent of K_d per unit base arclength and, for $d \geq 2$, uses no derivative of the base.

Lemma 3.10 (turning exclusion). *[proved here] Let γ be a C^1 arc from X_1 to X_2 with unit tangent T_γ , T_i its tangent at X_i , $a(\sigma) = \angle(T_\gamma(\sigma), T_1)$ and $b(\sigma) = \angle(T_\gamma(\sigma), T_2)$. If $a + b < \pi$ along γ , then $(X_2 - X_1) \cdot (T_1 + T_2) > 0$; in particular (X_1, X_2) is not doubly critical. This holds whenever the total turning $\int_\gamma |T'_\gamma|$ is $< \pi$.*

Proof. As $a, b \geq 0$ and $a + b < \pi$, $\cos a + \cos b = 2 \cos \frac{a+b}{2} \cos \frac{a-b}{2} > 0$, so $(X_2 - X_1) \cdot (T_1 + T_2) = \int_\gamma T_\gamma \cdot (T_1 + T_2) d\sigma > 0$, while a doubly critical pair has $(X_2 - X_1) \cdot T_1 = (X_2 - X_1) \cdot T_2 = 0$. The last sentence follows from $a(\sigma) \leq \int_0^\sigma |T'_\gamma|$ and $b(\sigma) \leq \int_\sigma^\ell |T'_\gamma|$. $\square$

In the notation of the proof of Proposition 3.8, $K'_d = v(1-x)T + rmW$ with $x = r\kappa_\perp \in [-f, f]$, so the angle $\chi \in (0, \pi/2)$ between K'_d and T has $\tan \chi = rm/(v(1-x))$ and $\chi_{\text{lo}} := \arctan \frac{rm}{v_{\max}(1+f)} \leq \chi \leq \chi_{\text{hi}} := \arctan \frac{rm}{v_{\min}(1-f)} = \arctan \frac{1}{\zeta_{\min}}$.

Proposition 3.11 (pairs on the same strand, $p = 2$, sharpened). *[proved here] Let $p = 2$, q odd, $f = r/\tau \leq \frac{1}{2}$, $m > 0$. Set $\delta_{R2} := (\pi - 2(\chi_{\text{hi}} - \chi_{\text{lo}}))/(1 + \sin \chi_{\text{hi}}/\Theta_{\min})$; if (H₃) of Proposition 3.8 holds, set $\delta_{R1} := \pi/(\tau\bar{\Omega})$ with*

$$\tau\bar{\Omega} := 1 + \sin^2 \chi_{\text{hi}} + \frac{\sin \chi_{\text{hi}}}{\Theta_{\min}} + \tau \frac{rm[M_v(1+f)/v_{\min} + rM_\kappa]}{v_{\min}^2(1-f)^2 + r^2 m^2},$$

and otherwise $\delta_{R1} := 0$; let $\delta_{\text{turn}} = \max(\delta_{R1}, \delta_{R2})$. Then every doubly critical pair of K_d with distinct base points and strand index $k = 0$ relative to the shorter base arc has $\delta \geq \delta_{\text{turn}}$ and

$$|X_2 - X_1| \geq \min(2(\tau - r), 2c'_0 r), \quad c'_0 := \frac{1}{2r} \inf_{\delta_{\text{turn}} \leq \delta < \pi} \max(\Lambda(\delta), \ell_0(\delta) - 2r).$$

In particular $c'_0 \geq \frac{1}{2}$ whenever $\delta_{\text{turn}} \geq \pi/3$.

Proof. If $\delta \geq \pi$ or the base chord is $\geq 2\tau$, Step 1 of Lemma 3.7 and Proposition 3.6(b) give $|X_2 - X_1| \geq 2(\tau - r)$. Otherwise let γ be the arc of K_d over the shorter base arc, $t \in [t_1, t_2]$, so $u = m(t_2 - t_1) \leq \delta/\Theta_{\min}$; since $k = 0$, γ joins X_1 to X_2 .

$\delta \geq \delta_{R2}$, without (H_3) . Put $\chi_i = \chi(t_i)$ and $Y_i(t) = \cos \chi_i T(t) + \sin \chi_i W(t)$. From $U' = mW - v\kappa_{\perp} T$, $W' = -mU - v\kappa_W T$, $T' = v(\kappa_{\perp} U + \kappa_W W)$ we get $Y'_i = (v\kappa_{\perp} \cos \chi_i - m \sin \chi_i)U + v\kappa_W \cos \chi_i W - v\kappa_W \sin \chi_i T$ and $|Y'_i|^2 = (v\kappa_{\perp} \cos \chi_i - m \sin \chi_i)^2 + v^2 \kappa_W^2 \leq (v\kappa + m \sin \chi_i)^2$, using $\kappa_{\perp}^2 \cos^2 \chi_i + \kappa_W^2 \leq \kappa^2$ and $-\kappa_{\perp} \cos \chi_i \leq \kappa$. The unit tangent $T_d(t) = \cos \chi(t) T(t) + \sin \chi(t) W(t)$ and $Y_i(t)$ lie in $\text{span}(T(t), W(t))$ at angles $\chi(t)$, χ_i from $T(t)$, and $Y_i(t_i) = T_d(t_i)$, so $a(t) \leq |\chi(t) - \chi_1| + \int_{t_1}^t (v\kappa + m \sin \chi_1)$ and $b(t) \leq |\chi(t) - \chi_2| + \int_t^{t_2} (v\kappa + m \sin \chi_2)$. With $\int v\kappa dt \leq \sigma/\tau = \delta$, $a + b \leq 2(\chi_{\text{hi}} - \chi_{\text{lo}}) + \delta + u \sin \chi_{\text{hi}} \leq 2(\chi_{\text{hi}} - \chi_{\text{lo}}) + \delta(1 + \sin \chi_{\text{hi}}/\Theta_{\min})$, which is $< \pi$ if $\delta < \delta_{R2}$; Lemma 3.10 excludes the pair.

$\delta \geq \delta_{R1}$, under (H_3) . The turning rate of T_d per unit t is $\omega = |K'_d \times K''_d|/|K'_d|^2$. The Minkowski split in the proof of Proposition 3.8, multiplied by $|K'_d| = \sqrt{A+B}$, gives $\omega \leq v\kappa(A+2B)/(A+B) + rm^2/\sqrt{A+B} + rm|a_0|/(A+B)$, where $B/(A+B) = \sin^2 \chi$ and $rm/\sqrt{A+B} = \sin \chi$; with $\kappa \leq 1/\tau$, $\chi \leq \chi_{\text{hi}}$, $v \geq v_{\min}$ and $|a_0| \leq M_v(1+f) + rvM_{\kappa}$ this is $\omega/v \leq \bar{\Omega}$ per unit base arclength, so the total turning of γ is $\leq \tau \bar{\Omega} \delta$, which is $< \pi$ if $\delta < \delta_{R1}$.

Distance. For $\delta < \pi$, Step 3 of Lemma 3.7, which does not use the strand index, gives $|X_2 - X_1| \geq \Lambda(\delta)$ and $|X_2 - X_1| \geq \ell_0(\delta) - 2r$; hence c'_0 . Finally $\Lambda \geq r$ on $[\pi/3, \pi)$ when $f \leq \frac{1}{2}$: on $[\pi/3, \pi/2]$, $\Lambda = 2(\tau - r) \sin(\delta/2) \geq \tau - r \geq r$; on $[\pi/2, \pi)$ this is Step 4 of Lemma 3.7, and Λ/r decreases in f . $\square$

Corollary 3.12 (same strand in the default family). *[proved here] Let $(p_j, q_j) = (2, 3)$ and $f_j = r_j/\text{thick}(K_{j-1}) \leq \frac{1}{2}$ for all $j \leq d$. Then every doubly critical pair of K_d with distinct base points on the same strand is at distance $\geq \min(2(\tau - r), 2c'_0 r)$ with $c'_0 \geq \frac{1}{2}$, at every depth $d \geq 1$, without (H_3) . At $d = 1$ and $f = \frac{1}{2}$, $\delta_{\text{turn}} \geq \delta_{R1} = \pi/(1 + \frac{9}{13} + \frac{9}{2\sqrt{13}}) > 1.0684$ and $c'_0 \geq \sin(\delta_{R1}/2) > 0.509$ (larger for smaller f); for $d \geq 2$, $\delta_{\text{turn}} \geq \delta_{R2} > 1.677$.*

Proof. $d = 1$. The base is the unit circle: $v \equiv 1$, $\kappa \equiv 1 = 1/\tau$, $\alpha = 0$, $m = \frac{3}{2}$, $\Theta_{\min} = \frac{2}{3}$, and (H_3) holds with $M_v = M_{\kappa} = 0$. With $\zeta = \zeta_{\min} = 2(1-f)/(3f)$, $\tau \bar{\Omega} = 1 + \frac{1}{1+\zeta^2} + \frac{3}{2\sqrt{1+\zeta^2}}$, which is $< 2.9404 < 3$ at $f = \frac{1}{2}$ ($\zeta = \frac{2}{3}$) and decreases as f decreases; so $\delta_{R1} > \pi/3$. On $[\delta_{R1}, \pi/2]$, $\Lambda/r = 2(1/f - 1) \sin(\delta/2) \geq c_* := 2 \sin(\delta_{R1}/2)$. On $[\pi/2, \pi)$ the argument of Step 4 of Lemma 3.7 with 1 replaced by c_* goes through: $\phi_{c_*}(\zeta) = 4\zeta^2 - c_*\zeta - 1 + \pi/2 - 2 \arcsin \zeta$ has the same second derivative, $\phi'_{c_*}(1/\sqrt{2}) = 2\sqrt{2} - c_* > 1.81$, $\phi_{c_*}(1/\sqrt{2}) = 1 - c_*/\sqrt{2} > 0.279$ and $\phi_{c_*}(1) = 3 - c_* - \pi/2 > 0.410$. Both δ_{R1} and Λ/r increase as f decreases.

$d \geq 2$. By (5), $|K'_1|^2 = (1-x)^2 + r_1^2 m_1^2$ with $r_1 = f_1$, $m_1 = \frac{3}{2}$, so after the reparametrisation $v_{\min}(K_1) \geq 2\sqrt{(1-f_1)^2 + \frac{9}{4}f_1^2}$, while $\text{thick}(K_1) \leq f_1$ (Proposition 3.5); hence $v_{\min}(K_1)/\text{thick}(K_1) \geq 2\sqrt{(1-f_1)^2/f_1^2 + \frac{9}{4}} \geq \sqrt{13}$. For $j \geq 2$, $v_{\min}(K_j) \geq 2(1-f_j)v_{\min}(K_{j-1})$ and $\text{thick}(K_j) \leq f_j \text{thick}(K_{j-1})$, so $v_{\min}/\text{thick}$ at least doubles per depth. With $m_d \leq \frac{3}{2} + \frac{1}{2} = 2$, $\Theta_{\min} \geq \sqrt{13}2^{d-3} \geq \sqrt{13}/2 > 1.802$ and $\zeta_{\min} = (1-f_d)\Theta_{\min}/f_d \geq \Theta_{\min}$. Dropping $\chi_{\text{lo}} \geq 0$, $\delta_{R2} \geq (\pi - 2 \arctan(1/\zeta_{\min}))/((1+1/(\Theta_{\min}\sqrt{1+\zeta_{\min}^2})))$, which increases in $\Theta_{\min}$ and $\zeta_{\min}$ and at $\Theta_{\min} = \zeta_{\min} = \sqrt{13}/2$ exceeds $1.677 > \pi/3$. (The bound $\Theta_{\min} \geq 2^{d-2}$ of Step 5 of Lemma 3.7 would not suffice at $d = 2$: at $f = \frac{1}{2}$, $\delta_{R2} \geq \pi/3$ needs $\Theta_{\min} \geq 1.12$.) Proposition 3.11 concludes. $\square$

Proposition 3.13 (curvature of K_d with the strand angle kept). *[proved here, under (H_3)] Under the hypotheses of Proposition 3.8, $\min \text{Rad}(K_d) \geq 1/\hat{\kappa}_d$, where, with $A = v^2(1-x)^2$, $B = r^2 m^2$ and $b_U = v^2(1-x)x/r - rm^2$,*

$$\hat{\kappa}_d := \sup_{|x| \leq f, v_{\min} \leq v \leq v_{\max}} \frac{[b_U^2(A+B) + (rm(M_v(1-x) + rvM_{\kappa}) + v(A+2B)\sqrt{\tau^{-2} - x^2/r^2})^2]^{1/2}}{(A+B)^{3/2}}.$$

Proof. By the identity in the proof of Proposition 3.8, $\kappa_{K_d}^2 = [b_U^2(A+B) + (rma_0 - \kappa_W v(A+2B))^2]/(A+B)^3$ with $x = r\kappa_{\perp}$, $|a_0| \leq M_v(1-x) + rvM_{\kappa}$ and $|\kappa_W| = (\kappa^2 - \kappa_{\perp}^2)^{1/2} \leq (\tau^{-2} - x^2/r^2)^{1/2}$; for fixed (x, v) the right-hand side is non-decreasing in $|a_0|$ and $|\kappa_W|$. $\square$

Theorem 3.14 (thickness lower bound for $p = 2$). [proved here; (b) under (H_3) ; the values in Table 1 are grid evaluations, not certified] Let $p = 2$, q odd, $f = r/\tau \leq 1/2$, $m > 0$. (a) $\frac{1}{2} \text{dcsd}(K_d) \geq \min(\tau - r, \min(c_1, c'_0) r)$, with $c_1 = c_1(f, \Theta_{\min})$ of Lemma 3.7 and c'_0 of Proposition 3.11. (b) Under (H_3) , $\min\text{Rad}(K_d) \geq 1/\hat{\kappa}_d$, hence $\text{thick}(K_d) \geq \min(\tau - r, cr)$ with $c = \min(c_1, c'_0, 1/(r\hat{\kappa}_d))$. (c) For the default family $(2, 3)$ with $f_j \leq \frac{1}{2}$ at every level, $\frac{1}{2} \text{dcsd}(K_d) \geq \min(\tau - r, r/2) = r/2$ and $\text{thick}(K_d) \geq \min(\min\text{Rad}(K_d), r/2)$ at every depth, without (H_3) ; at $d = 1$, $\text{thick}(K_1) \geq r_1/2$.

Proof. By (1), $\text{thick} = \min(\min\text{Rad}, \frac{1}{2} \text{dcsd})$. Doubly critical pairs with the same base point are at distance $2r$ (Proposition 3.6(a)); if the shorter base arc has $\delta \geq \pi$, the base chord is $\geq 2\tau$ (Step 1 of Lemma 3.7) and the distance is $\geq 2(\tau - r)$ (Proposition 3.6(b)); pairs with $\delta < \pi$ are covered by Lemma 3.7 ($k = 1$) and Proposition 3.11 ($k = 0$). Since $c_1 \leq 1$ (its limit as $\delta \rightarrow 0$ is 1), this gives (a); (b) is Proposition 3.13. For (c), $c_1 \geq \frac{1}{2}$ is Lemma 3.7, $c'_0 \geq \frac{1}{2}$ is Corollary 3.12, and $\tau - r \geq r$. At $d = 1$ ($M_v = M_\kappa = 0$, $v \equiv 1$), Proposition 3.8 gives $r\bar{\kappa}_1 = fG(\zeta_{\min})/(1 - f) + 1/(1 + \zeta_{\min}^2)$ with $\zeta_{\min} = 2(1 - f)/(3f)$ and $G \leq \max g = g(\sqrt{2}) = 4\sqrt{2}/(3\sqrt{3})$ (the derivative of $y(y + 2)^2/(y + 1)^3$, $y = \zeta^2$, is proportional to $(y + 2)(2 - y)$); this increases with f , and at $f = \frac{1}{2}$ equals $g(\sqrt{2}) + \frac{9}{13}$, so $1/(r\bar{\kappa}_1) > 0.561 > \frac{1}{2}$. $\square$

Without the two refinements of this version, Propositions 3.8–3.9 give at $f = 1/2$, $d = 1, 2, 3$ only $c_0 = 0.32, 0.13, 0.16$ and $1/(r\bar{\kappa}_d) = 0.56, 0.49, 0.85$. Table 1 lists the refined values on the polygons; in every measured level $c \geq 0.509$ and the minimum is attained by the different-strand constant c_1 except at $(f = \frac{1}{2}, d = 1)$, where it is c'_0 (with the term P_0 of Proposition 3.9 restricted to $\delta \geq \delta_{\text{turn}}$, c'_0 rises there to 0.650, a grid value).

Table 1: Constants of Theorem 3.14 on the polygons ($N_0 = 256$; τ, v, α and, for $1/(r\hat{\kappa}_d)$ at $d \geq 2$, M_v, M_κ measured on the base polygon; $\delta_{\text{turn}} = \delta_{R1}$ at $d = 1$, δ_{R2} at $d \geq 2$), rounded down. Grid evaluations, not certified (experiments/check_Hc.py, check_same_strand.py); analytic: $c_1, c'_0 \geq \frac{1}{2}$ in every row. At $f = 0.25$ all three are ≥ 0.999 . Last column: measured τ_d/r_d .

f	d	c_1	c'_0 (δ_{turn})	$1/(r\hat{\kappa}_d)$	c	τ_d/r_d
0.35	1	0.965	1.156 (1.344)	2.215	0.965	1.000
0.35	2	0.999	1.824 (2.838)	1.878	0.999	1.000
0.35	3	0.999	1.856 (3.107)	1.855	0.999	1.000
0.5	1	0.611	0.509 (1.068)	1.419	0.509	0.832
0.5	2	0.699	0.845 (2.056)	1.034	0.699	1.000
0.5	3	0.706	0.988 (2.928)	0.998	0.706	1.000

Remark 3.15 (no constant works for an arbitrary base). [proved here] No $c > 0$ depending only on (p, q, f) makes $\text{thick}(K_d) \geq \min(\tau - r, cr \sin(\pi/p))$ true for every smooth base K with $\text{thick}(K) = \tau$; the hypothesis of v0.1, stated for an arbitrary K , was therefore false (two counterexamples found in the first review). (i) If the curvature of K rises from 0 to $1/\tau$ over an arc of length ε (a smoothed junction of a segment and a circle of radius τ), by (5) the tangent of the inner strand turns from $vT + rmW$ to $v(1 - f)T + rmW$ over that arc, so $\kappa_{K_d} \gtrsim rv\kappa'/|K'_d|^2 \rightarrow \infty$: the M_v, M_κ term of (9) is unavoidable. (ii) If K is traversed slowly ($v \rightarrow 0$) on an arc of length $\gg r$, the two strands there are helices of pitch $h = v/m \rightarrow 0$; in the straight-tube model $|K_d(t) - K_d(t + 2\pi + u)|^2 = 4r^2 \cos^2(qu/4) + v^2 u^2$, whose minimum over u is $\leq (2\pi v/q)^2$ and is attained at a doubly critical pair by screw symmetry, so $\text{thick}(K_d) \leq \pi v/q \rightarrow 0$ (in Lemma 3.7, $c_1 \rightarrow 0$ as $\Theta_{\min} \rightarrow 0$). In the family $\Theta_{\min}$ is bounded below a priori (Lemma 3.7, Corollary 3.12); no a priori bound on M_v, M_κ uniform in d is known.

What remains unproved is therefore a statement about the family, not about an arbitrary curve:

Hypothesis 3.16 (H_c , for the family). [for $(2, 3)$ and $c = 1/2$: proved at $d = 1$, reduced to a curvature bound for $d \geq 2$ (Theorem 3.14); verified numerically] Let (K_d) be the family of Definition 2.1 built from the circle with a fixed pattern (p, q) , and let $f \leq 1/2$. There is $c = c(p, q, f) \in (0, 1]$ such that, for every sequence (r_d) with $r_d \leq f \text{thick}(K_{d-1})$ and every $d \geq 1$,

$$\text{thick}(K_d) \geq \min(\text{thick}(K_{d-1}) - r_d, cr_d \sin(\pi/p)). \quad (10)$$

Status for $(2, 3)$, $f \leq 1/2$, $c = 1/2$. By Theorem 3.14(c), (10) holds at $d = 1$ and, for $d \geq 2$, is equivalent to $\min\text{Rad}(K_d) \geq r_d/2$. Under (H_3) this follows from $1/(r\hat{\kappa}_d) \geq 1/2$, which the grid evaluation with polygon-measured inputs gives at $d = 2, 3$ (1.034, 0.998 at $f = 1/2$; Table 1); this is evidence, not a proof,

since neither $\text{thick}(K_{d-1})$ nor M_v, M_κ is known a priori. Open: an a priori bound for (H_3) uniform in d , and certification. Numerically (10) with $c = 1/2$ holds in all 9 default levels with margin $\tau_d/\rho_d \geq 1.66$, and along the chain of Corollary 3.17 with margin ≥ 1.66 .

On the constant. The natural constant is $c = 1$: $r \sin(\pi/p)$ is half the strand chord, and by Propositions 3.5 and 3.6(a) it cannot be improved where $\tau_d = r_d$. With $c = 1$ the inequality fails on the polygons in 4 of the 15 measured levels with $f \leq 1/2$ $((2, 3), f = 0.5, d = 1 (0.832); (3, 2), d = 1 (0.968); (3, 2), d = 2 (0.999); (2, 5), d = 1 (0.564)$); the deficit at $(3, 2), d = 2$ is stable under doubling of N ; it holds with equality, to 0.05%, in every level with $\tau_d = r_d$ and in $(3, 2), d = 3; (2, 5), d = 2, 3$, where the critical pair is a pair of adjacent strands of one disc. The value $c = 1/2$ was chosen after $c = 1$ had failed (Section 4); Theorem 3.14 reaches it, while $\tau_1/r_1 = 0.832$ at $f = 1/2$ shows that no argument reaches $c = 1$ there.

Corollary 3.17 (recursion for the ropelength bound). *[proved here, conditional on H_c for the family (for $(2, 3), c = 1/2$: on $\min\text{Rad}(K_d) \geq r_d/2, d \geq 2$)] Fix $(p, q), f \leq 1/2$ and assume (H_c) with constant c (for all $r_d \leq f \text{thick}(K_{d-1})$). Define $\rho_0 = 1, r_d := f\rho_{d-1}$ and $\rho_d := \gamma\rho_{d-1}$ with*

$$\gamma = \min(1 - f, cf \sin(\pi/p)), \quad A = p(1 + f) + f(q + \frac{p}{2}). \quad (11)$$

Then $\text{thick}(K_d) \geq \rho_d = \gamma^d$ and $L(K_d) \leq 2\pi A^d$ for all d , hence $\text{Rop}(K_d) \leq 2\pi\Lambda^d$ with $\Lambda = A/\gamma$. For $(2, 3)$ and $f = 1/2$: $A = 5.00$, and $\Lambda = 20.0$ with $c = 1/2$ ($\gamma = 0.250$), $\Lambda = 10.0$ with $c = 1$ ($\gamma = 0.500$).

Proof. Induction on d . If $\text{thick}(K_{d-1}) \geq \rho_{d-1}$ then $r_d = f\rho_{d-1} \leq f \text{thick}(K_{d-1})$, so (H_c) applies and $\text{thick}(K_d) \geq \min(\text{thick}(K_{d-1}) - r_d, cr_d \sin(\pi/p)) \geq \min(\rho_{d-1} - f\rho_{d-1}, cf\rho_{d-1} \sin(\pi/p)) = \gamma\rho_{d-1} = \rho_d$. By Lemma 3.3 with $\tau \geq \rho_{d-1}$, $L(K_d) \leq pL(K_{d-1})(1 + f) + f\rho_{d-1}(2\pi q + \pi p)$; since $\rho_{d-1} \leq \text{thick}(K_{d-1}) \leq L(K_{d-1})/2\pi$, this is $\leq L(K_{d-1})[p(1 + f) + f(q + p/2)] = AL(K_{d-1})$, and $L(K_0) = 2\pi$. $\square$

Remark 3.18 (on the constants). Λ is generous: Lemma 3.3 bounds $|K'_d|$ by the sum of the norms of two orthogonal vectors, uses $\kappa_{\max} \leq 1/\tau$ and $|\alpha| \leq \pi$, and (H_c) with $c = 1/2$ is twice weaker than the observed $\tau_d \approx r_d$. Numerically the growth factor per depth is 4.06, 4.00 at $f = 1/2$ (Table 2) against $\Lambda = 20.0$; the conditional bounds $2\pi\Lambda^d$ are 126, 2513, 50265; along the chain of the corollary itself ($r_d = f\rho_{d-1}$, `experiments/corollary_chain.py`) the polygons give $L/\tau = 38.4, 256.5, 2051.9$ and $\tau_d/\rho_d \geq 1.66$ for $f \in \{0.25, 0.35, 0.5\}$. We make no claim that any configuration in the family is stationary or optimal for ropelength, and none that the recursion is a “flow law” for tight recursive knots: it bounds one explicit construction.

Remark 3.19 (lower bounds, quoted). [literature] Buck and Simon [2] prove $\text{Rop}(K) \geq c_{\text{BS}} \text{Cr}(K)^{3/4}$ for a universal $c_{\text{BS}} > 0$, and Cantarella, Kusner and Sullivan [3] prove that ropelength minimisers exist in every knot and link type, are $C^{1,1}$, and satisfy further lower bounds. These need $\text{Cr}(K_d)$. K_2 is a 2-cable of the trefoil (Lemma 3.2), which is alternating, hence adequate; Kalfagianni and McConkey [8] prove that a cable with q strands of an adequate knot K has crossing number $\geq q^2 \text{Cr}(K) + 1$, so $\text{Cr}(K_2) \geq 13$, and they determine the crossing numbers of 2-cables of adequate knots (we have not matched their cable and writhe conventions with Lemma 3.2, so we do not quote the exact value). For $d \geq 3$ only general satellite bounds are available, such as $\text{Cr} \geq 10^{-13} \text{Cr}(\text{companion})$ [9]. For $K_1 = T(2, 3)$ the best known numerical ropelength is about 32.7 (numerical, uncertified; [3, 12]); our depth-one polygon at $f = 1/2$ gives 38.4, i.e. the construction is not close to tight even at depth one.

Conjecture 3.20 (a threshold for $\text{thick}(K_d) = r_d$). [conjectural] For the default family there are thresholds $f_*(d) > 0$ with $\inf_d f_*(d) > 0$ such that for $f \leq f_*(d)$ every doubly critical pair of K_d other than the antipodal pairs of one normal disc is at distance $> 2r_d$ and $\min\text{Rad}(K_d) > r_d$; hence $\text{thick}(K_d) = r_d$ (the inequality $\leq$ is Proposition 3.5). Numerically the largest admissible $f_*(1)$ lies in $[0.35, 0.5]$, and $f_*(2), f_*(3) \geq 0.5$, the latter by margins of only 1.4% and 2.1% (Section 4).

Remark 3.21 (growth rate). If $\text{thick}(K_d) = r_d$ for all large d and $\min\text{Rad}(K_{d-1})$ stays bounded below, then $\text{Rop}_d/\text{Rop}_{d-1} = (L_d/L_{d-1})(\tau_{d-1}/\tau_d) \rightarrow p/f$ by Lemma 3.3. The measured ratios at $d = 2, 3$ are 8.01, 8.00 ($f = 0.25; p/f = 8.00$), 5.73, 5.71 (5.71), 4.06, 4.00 (4.00), 4.03, 4.00 ((2, 5); 4.00) and 6.95, 6.93 ((3, 2), adjacent strands, limit $p/(f \sin(\pi/p)) = 6.93$): the rate is a consequence of the conjecture, not an independent constant.

4 Experiments

Criteria. The protocol is in Appendix A. Four criteria are checked: (C1) resolution: Rop and τ change by less than 1% between N_0 and $2N_0$; (C2) $L(K_d) \leq$ the right-hand side of (6) evaluated with the

measured $L(K_{d-1})$, $\tau(K_{d-1})$ and α_d ; (C3) $\tau(K_d) \geq \rho_d$ of (10) with $c = 1/2$ and the measured $\tau(K_{d-1})$, for $f \leq 1/2$; (C4) τ_{pt} agrees with τ to within 1%. Which of these were fixed in advance: the plan written before any code (`PLAN.md`) fixed the resolution check at $d = 1, 2$ without a threshold, the length bound and the form of (10) with an unspecified constant; the 1% thresholds, the extension of (C1) to $d = 3$ and the cross-check (C4) were fixed when the manuscript was written, and the constant $c = 1/2$ of (C3) was fixed *after* a first run with the natural constant $c = 1$ had failed at $(f = 1/2, d = 1)$. (C3) is therefore a post hoc check, reported as such.

Results. Table 2 gives all patterns at the finest resolution and Figure 1 the growth (projections of the curves: `figures/fig_curves.pdf`). All four criteria hold: (C1) the maximal relative change is 0.03% for Rop and 0.03% for τ over all 12 chains (per-level values in `results/tables.md`); (C2) holds in 36 of 36 levels; (C3) holds in all 9 default levels with $f \leq 1/2$, with margin $\tau_d/\rho_d \geq 1.66$ (and ≥ 1.13 including the other patterns); with $c = 1$ instead of $1/2$ the margin drops to 0.56 and the inequality fails in 4 of 15 levels, as discussed after Hypothesis 3.16; (C4) the maximal discrepancy is 0.01%. In every level τ is determined by $\text{dcsd}/2$, never by minRad (36 of 36; $\text{minRad}/r_d \geq 1.15$ throughout).

Three further observations. First, for $f \leq 0.35$ the measured $\tau(K_d)$ equals r_d to four digits at every depth (the antipodal strands in a normal disc are the doubly critical pair, consistent with Proposition 3.5 and Conjecture 3.20), so $\text{Rop}_d = L_d/r_d$ and the recursion $\tau_d = f\tau_{d-1}$ is exact; for $f = 1/2$ at depth one $\tau_1 = 0.4158 < r_1 = 0.5$, realised by the pair across the hole of the torus described in Section 3, while at depths 2, 3 again $\tau_d = r_d$, but narrowly: the next-closest doubly critical vertex pairs (base points at arc $\geq \pi\tau$) are only 1.4% and 2.1% farther than $2r_d$ (1.4% and 2.0% at $N_0 = 512$; 103% and 109% at $f = 0.35$), so $f = 1/2$ is close to the threshold of Conjecture 3.20. In the same census (`experiments/classify_pairs.py`, 18 levels of the default family, $f \leq 1/2$, $N_0 \in \{256, 512\}$) no doubly critical vertex pair on the same strand with distinct base points occurs; a sign-change test that also detects saddles and maxima of the distance finds same-strand doubly critical pairs only at $d = 1$, with $\delta \geq 1.96 > \delta_{\text{turn}}$ and distance $\geq 3.70 r_1$ at $f = 1/2$, and none at $d = 2, 3$. Second, the writhe of the depth-one configuration crosses the half-integer 3.5 inside the grid ($\text{Wr}(K_1) = 3.127, 3.257, 3.518, 3.710$ for $f = 0.25, 0.35, 0.5, 0.6$; the crossing is at $r_1^* = 0.4904$ by bisection), so by Lemma 3.2 the frame linking number is 3 for $f \leq 0.35$ and 4 for $f \geq 0.5$ and the depth-two knot type changes across the grid: K_2 is the $(2, 9)$ -cable of the trefoil for $f = 0.25$ but the $(2, 11)$ -cable for $f = 0.5$. The frame linking number is computed as $\text{rint}(\text{Wr } K_{d-1} - \alpha_d/2\pi)$; the residual $|\text{Wr } K_{d-1} - \alpha_d/2\pi - n|$ is a consistency check of Lemma 3.2 and is at most $1.1 \cdot 10^{-3}$ at the fine resolution ($4.5 \cdot 10^{-3}$ at the coarse one) over all levels. Third, the best depth-one value on the grid is $f = 0.5$ ($\text{Rop}_1 = 38.4$), the best depth-three value also ($\text{Rop}_3 = 622.7$); $f = 0.6$ is worse at every depth, but we did not optimise f and make no optimality claim.

Writhe accuracy. The midpoint-rule writhe changes under doubling of N by at most $1 \cdot 10^{-4}$ at depth one of the default family, but by up to 0.031 at depth three of $(3, 2)$ (the double sum converges slowly on the long chains); the knot-type decisions are unaffected (the smallest margin to a half-integer is 0.018 at $f = 1/2$, $d = 1$, where the change is $\leq 10^{-4}$), but the accuracy claimed for Wr is level-dependent. An exact polygonal writhe (Levitt; Klenin–Langowski) would remove this at the same cost and is listed as a next step.

Curve-shortening probe (uncertified). 200 explicit discrete curve-shortening steps ($\lambda = 0.1$, `experiments/shrink.py`) lower L/τ of K_1 ($f = 1/2$, $N_0 = 512$) from 38.4 to 38.1 and raise that of K_2 from 155.6 to 156.3: no evidence about local minimality (`results/tables_shrink.md`).

Table 2: All chains at the finest resolution ($N_0 = 1024$; 512 for $(3, 2)$). $r_d = f\tau_{d-1}$; $\rho_d^{(1)} = \min(\tau_{d-1} - r_d, r_d \sin(\pi/p))$ is (10) with the natural constant $c = 1$, so $\tau_d/\rho_d^{(1)} < 1$ marks a failure of $c = 1$ (the ratio for $c = 1/2$ is twice as large when $\tau_{d-1} - r_d$ is not the minimum); slope = $q + pn$ of Lemma 3.2. † : $f = 0.6$ lies outside the hypothesis $r \leq \tau/2$ and is a numerical probe only. Lengths in units of R_0 .

(p, q)	f	d	N	r_d	L	τ	L/τ	$\tau_d/\rho_d^{(1)}$	Wr	slope
(2, 3)	0.25	1	2048	0.2500	13.444	0.2500	53.8	1.000	3.13	3
(2, 3)	0.25	2	4096	0.0625	26.911	0.0625	430.6	1.000	15.25	9
(2, 3)	0.25	3	8192	0.0156	53.822	0.0156	3444.6	1.000	63.50	33
(2, 3)	0.35	1	2048	0.3500	14.271	0.3500	40.8	1.000	3.26	3
(2, 3)	0.35	2	4096	0.1225	28.608	0.1225	233.5	1.000	15.49	9
(2, 3)	0.35	3	8192	0.0429	57.219	0.0429	1334.6	1.000	63.98	33
(2, 3)	0.5	1	2048	0.5000	15.949	0.4158	38.4	0.832	3.52	3
(2, 3)	0.5	2	4096	0.2079	32.349	0.2079	155.6	1.000	17.99	11
(2, 3)	0.5	3	8192	0.1040	64.730	0.1040	622.7	1.000	74.96	39
(2, 3)	0.6 †	1	2048	0.6000	17.307	0.3401	50.9	0.850	3.71	3
(2, 3)	0.6 †	2	4096	0.2041	34.941	0.1367	255.6	1.005	18.36	11
(2, 3)	0.6 †	3	8192	0.0820	69.893	0.0820	852.1	1.500	75.70	39
(3, 2)	0.5	1	1536	0.5000	20.000	0.4191	47.7	0.968	3.90	2
(3, 2)	0.5	2	4608	0.2096	60.087	0.1813	331.4	0.999	39.66	14
(3, 2)	0.5	3	13824	0.0906	180.270	0.0785	2296.5	1.000	362.97	122
(2, 5)	0.5	1	2048	0.5000	20.420	0.2822	72.4	0.564	6.87	5
(2, 5)	0.5	2	4096	0.1411	41.116	0.1411	291.4	1.000	32.66	19
(2, 5)	0.5	3	8192	0.0705	82.273	0.0705	1166.2	1.000	136.30	71

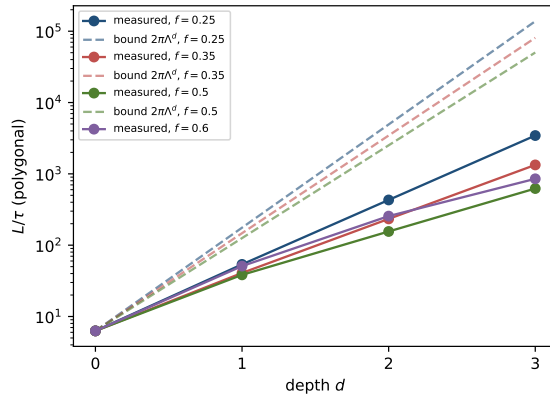


Figure 1: Polygonal L/τ against depth for the default family (solid) and the conditional bound $2\pi\Lambda^d$ with $c = 1/2$ (dashed, $f \leq 1/2$).

5 Claims and their status

Claim	Status
K_d is embedded, smooth and regular when $r_d < \text{thick}(K_{d-1})$ (Lemma 3.1).	proved here
The closed Bishop frame has $\text{Lk} = \text{Wr} - \alpha/2\pi$; K_d is the $(p, q + pn)$ -cable of K_{d-1} (Lemma 3.2).	proved here from classical results
$pL(K_{d-1})(1 - r\kappa_{\max}) \leq L(K_d) \leq pL(K_{d-1})(1 + r\kappa_{\max}) + r(2\pi q + p \alpha)$ (Lemma 3.3).	proved here
$\text{thick}(K_d) \leq r_d$ and $\text{Rop}(K_d) \geq L(K_d)/r_d \geq 2\pi(p(1-f)/f)^d$ for $p = 2$ (Prop. 3.5).	proved here (lower bound internal to the family)
Same-disc pairs at distance $\geq 2r \sin(\pi/p)$; far pairs at distance $\geq 2(\tau - r)$ (Prop. 3.6).	proved here
Different-strand pairs at distance $\geq \min(2(\tau - r), 2c_1 r)$; $c_1 \geq 1/2$ in the $(2, 3)$ family, $f \leq 1/2$, all d (Lemma 3.7).	proved here ($c_1 \geq 1/2$); grid value 0.6117 uncertified
Same-strand doubly critical pairs at distance $\geq \min(2(\tau - r), 2c'_0 r)$; $c'_0 \geq 1/2$ in the $(2, 3)$ family, $f \leq 1/2$, all d , no (H_3) (Prop. 3.11, Cor. 3.12).	proved here (round 2); refined values are grid evaluations
$\text{minRad}(K_d) \geq 1/\bar{\kappa}_d, \geq 1/\hat{\kappa}_d$; same-strand pairs $\geq \min(2(\tau - r), 2c_0 r)$ (Props. 3.8, 3.9, 3.13).	proved here under (H_3) ; unconditional at $d = 1$
$\frac{1}{2} \text{dcsd}(K_d) \geq \min(\tau - r, r/2)$ for $(2, 3)$, $f \leq 1/2$, all d ; $\text{thick}(K_1) \geq r_1/2$ (Thm. 3.14).	proved here
$\text{thick}(K_d) \geq \min(\tau - r, cr)$, $c = 1/2$, at $d = 2, 3$; values of Table 1.	grid evaluation of a bound proved under (H_3) , inputs measured on polygons (uncertified)
No $c > 0$ depending only on (p, q, f) works for an arbitrary base K (Remark 3.15).	proved here
(H_c) for $(2, 3)$ with $c = 1/2$ at every depth: (H_3) uniformly in d .	open; verified numerically in 9 levels, margin ≥ 1.66 ($c = 1$ fails in 4 of 15 levels)
$\text{Rop}(K_d) \leq 2\pi\Lambda^d$, $\Lambda = A/\gamma$ (Cor. 3.17).	proved conditional on (H_c) for the family
Explicit polygons with $L/\tau = 38.4, 155.6, 622.7$ at $d = 1, 2, 3$ ($f = 0.5$), stable to 0.03%.	computationally verified (polygonal proxy, not certified smooth thickness)
Knot type of K_2 changes between $f = 0.35$ and $f = 0.5$ (writhe of K_1 crosses 3.5 at $r_1^* = 0.4904$).	computationally verified ($\text{Wr}(K_1)$ stable to 10^{-4}), consequence of Lemma 3.2
$\text{Rop} \geq c_{\text{BS}} \text{Cr}^{3/4}$; tight knots; $\text{Cr}(K_2) \geq 13$; satellite bounds.	literature [2, 3, 8, 9]
$\text{thick}(K_d) = r_d$ for $f \leq f_*(d)$ (Conj. 3.20); then $\text{Rop}_d / \text{Rop}_{d-1} \rightarrow p/f$.	conjectural; at $f = 1/2$, $d = 2, 3$ margins 1.4%, 2.1%
Curve-shortening lowers the polygonal L/τ of K_1 by 0.8% and raises that of K_2 .	uncertified numerical probe; negative at depth two
Stationarity or optimality of any K_d ; a recursive flow law for tight recursive knots.	not claimed

6 Limitations and next steps

Limitations. (1) The thickness lower bound ($p = 2$, default family) is proved for all doubly critical pairs at every depth and completely at $d = 1$; for $d \geq 2$ the curvature of K_d needs (H_3) , whose constants are only measured on the polygons and whose uniformity in d is open. (2) The polygonal τ of (4) uses vertex pairs and a local-minimum test; agreement with τ_{pt} to 0.01%, the segment–segment check of Section 2.4 and stability under doubling of N are evidence, not certification, which would require interval arithmetic on a $C^{1,1}$ interpolant. (3) r_d is a fixed fraction of the measured thickness; nothing is optimised, and the family (pattern angle linear in t , phase $\varphi_d = 0$) is one of many; $p \geq 3$ is not treated in Section 3 beyond Propositions 3.5–3.6. (4) Only the thresholds $c_1 \geq \frac{1}{2}$, $c'_0 \geq \frac{1}{2}$ and $1/(r\bar{\kappa}_1) > 0.561$ are analytic; every other constant of Section 3 is a grid infimum or supremum (c_1, c'_0 with a Lipschitz constant estimated on the grid itself, $\hat{\kappa}_d$ with a neighbour-increment correction, c_0 of Proposition 3.9 on a 1500×400 grid without correction), with polygon-measured inputs for $d \geq 2$; none uses interval arithmetic. (5) Writhe and holonomy are midpoint-rule sums, accurate to 10^{-4} at depth one but only to 0.031 at depth three of $(3, 2)$; the frame linking number for $f = 0.5$ is decided by a margin of 0.02. (6) The only lower bound specific to the family is Proposition 3.5; knot-type bounds are limited by what is known of $\text{Cr}(K_d)$ (Remark 3.19).

Next steps. (1) Bound M_v, M_κ of K_d in terms of those of K_{d-1} to make (H_3) uniform in d ; with Theorem 3.14 this would prove (H_c) with $c = 1/2$ for $(2, 3)$ at every depth, and Corollary 3.17 unconditionally. (2) Certify: interval arithmetic for $c_1, c'_0, \hat{\kappa}_d$ and for $\text{minRad}, \text{dcsd}$ of a piecewise-circular interpolant; exact polygonal writhe (Levitt; Klenin–Langowski). (3) Replace the curve-shortening probe by a constrained length minimisation at fixed $\tau \geq 1$ with a certified knot-type check, for $d = 1, 2$. (4) Optimise f_d per depth and the pattern angle in arclength; treat $p \geq 3$ (the bound δ_{R2} is general, the a

priori bounds use $m \leq 2$ and $m_1 = 3/2$). (5) Match the conventions of [8] with Lemma 3.2 to obtain $\text{Cr}(K_2)$ and attach a Buck–Simon bound to each depth-two configuration.

A Reproducibility and protocol

Protocol. `experiments/recursive_knots.py` builds the polygon K_0 with N_0 vertices, computes the closed Bishop frame by discrete parallel transport (rotation taking T_i to T_{i+1} , Rodrigues formula; initial normal orthogonal to T_0 ; holonomy measured as the signed angle between the transported and the initial normal, removed by the twist $-\alpha_i/N$) and produces K_d with $p_d N_{d-1}$ vertices by (3) evaluated at the vertices (no interpolation; the parameter t is the vertex index). At each depth it measures L , minRad , dcsd , τ , τ_{pt} , Rop , the writhe, the holonomy and $\text{Lk} = \text{rint}(\text{Wr } K_{d-1} - \alpha_d/2\pi)$ with its residual, and sets $r_{d+1} = f \tau(K_d)$. Grid: $(p, q) = (2, 3)$ with $f \in \{0.25, 0.35, 0.5, 0.6\}$, and $(3, 2)$, $(2, 5)$ with $f = 0.5$; depths 1–3; two resolutions, $N_0 \in \{512, 1024\}$ ($\{256, 512\}$ for $(3, 2)$, whose depth-3 polygon has $27N_0$ vertices). The value $f = 0.6$ lies outside the hypothesis $r \leq \tau/2$ and is included only as a numerical probe. The computation is deterministic (seed 20260930 recorded for bookkeeping only); the reference run took 184s of wall-clock time (3.1 minutes) on shared cores with Python 3.11.15 and NumPy 2.4.6. `experiments/aux_checks.py` computes the critical-pair geometry at depth one, the writhe crossing r_1^* and the segment–segment check (`results/aux_checks.json`); `experiments/check_Hc.py` and `check_same_strand.py` evaluate the constants of Section 3 on the polygons and check every intermediate inequality of the proofs on all local pairs; they are copies, frozen for v0.3, of the scripts in `theory/` where the derivations were done (only paths changed; outputs byte-identical), and their outputs `results/check_*_summary.json` carry SHA-256 hashes in `results/FROZEN_THEORY_SHA256.txt`. `experiments/classify_pairs.py` classifies the doubly critical pairs and `corollary_chain.py` measures the chain of Corollary 3.17. `python3 experiments/make_numbers.py` writes `manuscript/numbers.tex` and the table body from these JSON files, and `manuscript/build.sh` runs it and `latexmk`. Every number in this text is a macro from `numbers.tex`.

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